

MALTA LUQA, Malta

WMO: 165970

Lat: 35.857N

Lon: 14.478E

Elev: 300

StdP: 14.54

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	44.4	46.3	31.9	26.5	49.9	35.2	30.4	50.6	31.6	53.9	27.8	53.6	7.9	290	0.425

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.3	93.5	70.5	91.1	71.3	88.2	71.6	77.6	83.8	76.5	82.8	75.5	81.9	8.2	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.7	135.8	79.9	74.9	132.4	79.6	73.5	126.2	79.2	41.2	83.3	40.1	82.9	39.2	81.8	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
25.5	22.6	20.2	40.5	100.0	1.8	3.9	39.2	102.8	38.1	105.1	37.1	107.3	35.7	110.1		
			36.5	79.8	2.1	1.8	35.0	81.1	33.8	82.1	32.7	83.1	31.1	84.4		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	66.7	54.9	54.3	57.0	61.1	67.7	75.3	80.2	81.0	76.2	70.5	63.8	57.7	(6)
	DBStd	10.17	3.55	3.80	3.73	3.53	3.89	4.70	3.44	3.21	3.50	3.73	3.95	3.74	(7)
	HDD50	14	5	7	1	0	0	0	0	0	0	0	0	1	(8)
	HDD65	1300	314	299	250	123	16	0	0	0	0	4	68	226	(9)
	CDD50	6116	156	128	218	334	547	759	937	961	786	636	413	241	(10)
	CDD65	1928	0	0	2	8	98	309	472	496	336	175	31	1	(11)
	CDH74	14924	0	0	8	22	389	2303	4576	4960	2157	491	18	0	(12)
	CDH80	5031	0	0	2	2	63	765	1825	1894	443	37	0	0	(13)
(14) Wind	WSAvg	9.2	10.2	10.5	10.6	10.8	9.8	8.3	7.6	7.2	8.1	8.2	9.3	10.2	(14)
(15) Precipitation	PrecAvg	20.4	3.1	2.2	1.5	0.8	0.3	0.1	0.0	0.2	1.4	3.5	3.2	3.9	(15)
	PrecMax	32.5	7.7	6.7	3.9	2.7	1.7	1.4	0.0	5.1	8.2	10.0	16.2	9.2	(16)
	PrecMin	9.5	0.4	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.3	0.0	0.7	(17)
	PrecStd	6.6	2.1	1.6	0.9	0.7	0.4	0.3	0.0	0.9	1.8	2.4	3.6	2.5	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	65.5	66.5	71.8	77.1	86.1	96.4	98.6	98.3	90.0	84.1	76.3	69.4	(19)
		MCWB	59.3	58.7	58.9	61.5	64.7	68.1	70.8	70.6	72.8	72.2	68.3	62.2	(20)
	2%	DB	63.2	64.0	67.5	72.4	80.9	91.0	93.6	93.3	86.2	80.5	73.7	66.5	(21)
		MCWB	58.2	57.8	58.3	60.1	64.3	69.1	70.4	71.8	72.9	71.0	66.7	60.3	(22)
	5%	DB	62.2	62.3	64.7	69.9	78.4	87.6	91.1	90.6	84.2	78.6	71.8	64.8	(23)
		MCWB	57.5	56.9	57.6	59.7	63.4	68.5	70.7	72.5	72.9	70.1	65.6	59.3	(24)
	10%	DB	60.7	60.6	62.8	67.7	75.4	84.3	88.1	88.1	82.5	76.8	70.0	63.6	(25)
		MCWB	56.1	55.7	56.8	59.2	63.2	68.0	71.2	73.1	72.4	69.5	64.1	58.5	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.3	60.2	61.2	63.8	69.2	75.2	78.1	79.5	78.3	75.8	70.7	63.4	(27)
		MCDB	63.7	64.1	67.2	71.3	76.7	84.0	86.6	85.3	82.7	79.0	73.9	67.0	(28)
	2%	WB	59.3	58.8	59.8	62.5	67.5	73.4	76.5	78.1	76.8	73.9	68.3	61.7	(29)
		MCDB	62.4	62.2	64.8	68.9	75.1	81.2	83.8	84.3	81.6	77.3	72.3	65.1	(30)
	5%	WB	58.1	57.5	58.8	61.2	66.2	72.0	75.4	77.0	75.7	72.5	66.6	60.3	(31)
		MCDB	61.3	61.1	63.2	66.7	73.2	80.1	83.0	83.6	80.2	76.3	70.7	63.9	(32)
	10%	WB	56.8	56.3	57.7	60.3	65.0	70.7	74.4	75.9	74.5	70.9	65.1	59.1	(33)
		MCDB	60.1	59.8	61.7	65.3	71.9	79.1	82.1	82.8	79.4	75.3	69.5	62.9	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	9.5	10.2	11.1	11.9	13.9	15.2	15.7	14.3	11.9	11.0	9.9	9.2	(35)
		MCDBR	10.1	11.1	12.9	15.2	17.7	18.9	19.5	17.6	13.6	12.2	11.1	9.8	(36)
	5% WB	MCWBR	7.1	7.5	7.4	7.1	8.0	8.9	10.1	9.3	6.8	6.5	7.3	6.9	(37)
		MCWBR	9.3	10.5	11.7	12.6	14.7	16.3	16.2	14.7	12.3	11.0	10.5	9.2	(38)
(40) Clear-Sky Solar Irradiance	taub		0.354	0.375	0.427	0.466	0.467	0.443	0.451	0.453	0.451	0.424	0.388	0.360	(40)
		taud	2.445	2.369	2.200	2.109	2.150	2.262	2.252	2.255	2.263	2.308	2.370	2.434	(41)
	Ebn at Noon		265	272	266	261	262	268	264	262	255	252	249	254	(42)
		Edh at Noon	28	34	43	50	48	43	43	43	40	35	30	27	(43)
(44) All-Sky Solar Radiation	RadAvg		814	1097	1515	1909	2248	2453	2474	2189	1705	1233	877	722	(44)
	RadStd		52	78	85	91	106	97	41	82	65	72	59	50	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	+0.32	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	-0.49	-0.54	N/A	N/A	N/A	N/A

Nomenclature: See separate page